

Jenkins Task

Task Description:

Launch jenkins and explore creating projects and users.

Techstacks needs to be used :

- AWS EC2
- Jenkins

How do I submit my work?

- Push all your work files to GitHub (O/P screenshot images must).
- Submit your URLs in the portal.

Terms and Conditions?

- You agree to not share this confidential document with anyone.
- You agree to open-source your code (it may even look good on your profile!). Do not mention our company's name anywhere in the code.
- We will never use your source code under any circumstances for any commercial purposes; this is just a basic assessment task.

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Jenkins Setup and Exploration Task

Objective

To install Jenkins and explore project creation and user management.

Tools Used

- Ubuntu Linux (VirtualBox VM)
- Jenkins
- Java 17

Environment

- Local Ubuntu VM (used instead of AWS EC2 equivalent setup)

```
root@nagios:~# cat /etc/os-release
PRETTY_NAME="Ubuntu 25.04"
NAME="Ubuntu"
VERSION_ID="25.04"
VERSION="25.04 (Plucky Puffin)"
VERSION_CODENAME=plucky
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=plucky
LOGO=ubuntu-logo
root@nagios:~#
```

Jenkins Installation

Jenkins was installed using the .deb package method.

wget https://pkg.jenkins.io/debian-stable/binary/jenkins_2.440.3_all.deb

```
sudir@nagios:~/jenkins$ wget https://pkg.jenkins.io/debian-stable/binary/jenkins_2.440.3_all.deb
--2026-04-08 11:15:34-- https://pkg.jenkins.io/debian-stable/binary/jenkins_2.440.3_all.deb
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.194.133, 151.101.130.133, 151.101.66.133, ...
Connecting to pkg.jenkins.io (pkg.jenkins.io)[151.101.194.133]:443... connected.
HTTP request sent, awaiting response... 302 Moved Temporarily
Location: https://get.jenkins.io/debian-stable/jenkins_2.440.3_all.deb [following]
--2026-04-08 11:15:35-- https://get.jenkins.io/debian-stable/jenkins_2.440.3_all.deb
Resolving get.jenkins.io (get.jenkins.io)... 128.24.70.119, 2603:1030:408:3::2bd
Connecting to get.jenkins.io (get.jenkins.io)[128.24.70.119]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://mirror.bom2.albony.in/jenkins/debian-stable/jenkins_2.440.3_all.deb [following]
--2026-04-08 11:15:37-- https://mirror.bom2.albony.in/jenkins/debian-stable/jenkins_2.440.3_all.deb
Resolving mirror.bom2.albony.in (mirror.bom2.albony.in)... 123.253.156.54, 2403:44c0:2:4::2
Connecting to mirror.bom2.albony.in (mirror.bom2.albony.in)[123.253.156.54]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 91438472 (87M) [application/vnd.debian.binary-package]
Saving to: 'jenkins_2.440.3_all.deb'

jenkins_2.440.3_all.deb      100%[=====>] 87.20M  16.0MB/s  in 9.7s

2026-04-08 11:15:47 (8.96 MB/s) - 'jenkins_2.440.3_all.deb' saved [91438472/91438472]
```

```
sudo apt install ./jenkins_2.440.3_all.deb -y
```

```
sudir@nagios:~/jenkins$ sudo apt install ./jenkins_2.440.3_all.deb -y
Note, selecting 'jenkins' instead of './jenkins_2.440.3_all.deb'
The following package was automatically installed and is no longer required:
  libutempter0
Use 'sudo apt autoremove' to remove it.

Installing:
  jenkins

Installing dependencies:
  net-tools

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 210 kB / 91.6 MB
  Space needed: 94.4 MB / 2,308 MB available

Get:1 /home/sudir/jenkins/jenkins_2.440.3_all.deb jenkins all 2.440.3 [91.4 MB]
Get:2 http://in.archive.ubuntu.com/ubuntu plucky-updates/main amd64 net-tools amd64 2.10-1.lubuntul.25.04.4 [210 kB]
Fetched 210 kB in 1s (152 kB/s)
Selecting previously unselected package net-tools.
(Reading database ... 111259 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-1.lubuntul.25.04.4_amd64.deb ...
Unpacking net-tools (2.10-1.lubuntul.25.04.4) ...
Selecting previously unselected package jenkins.
Preparing to unpack .../jenkins_2.440.3_all.deb ...
Unpacking jenkins (2.440.3) ...
Setting up net-tools (2.10-1.lubuntul.25.04.4) ...
Setting up jenkins (2.440.3) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/jenkins.service' -> '/usr/lib/systemd/system/jenkins.service'.
Processing triggers for man-db (2.13.0-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.
```

Jenkins Service Status

```
sudo systemctl status jenkins
```

```
sudir@nagios:~/jenkins$ sudo systemctl start jenkins
sudir@nagios:~/jenkins$ sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
sudir@nagios:~/jenkins$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-04-08 11:16:48 UTC; 28s ago
  Invocation: 5dd97a70647c4386a1363ba91f06f566
    Main PID: 34219 (java)
      Tasks: 51 (limit: 3979)
     Memory: 1G (peak: 1G)
        CPU: 1min 29.500s
    CGroup: /system.slice/jenkins.service
            └─34219 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenk

Apr 08 11:16:09 nagios jenkins[34219]: 9f377cd91ccd4alca9f2300380533735
Apr 08 11:16:09 nagios jenkins[34219]: This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Apr 08 11:16:09 nagios jenkins[34219]: *****
Apr 08 11:16:09 nagios jenkins[34219]: *****
Apr 08 11:16:09 nagios jenkins[34219]: *****
Apr 08 11:16:48 nagios jenkins[34219]: 2026-04-08 11:16:48.597+0000 [id=36] INFO jenkins.InitReactorRunn
Apr 08 11:16:48 nagios jenkins[34219]: 2026-04-08 11:16:48.641+0000 [id=24] INFO hudson.lifecycle.Lifecyc
Apr 08 11:16:48 nagios systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Apr 08 11:16:51 nagios jenkins[34219]: 2026-04-08 11:16:51.002+0000 [id=51] INFO h.m.DownloadService$Down
Apr 08 11:16:51 nagios jenkins[34219]: 2026-04-08 11:16:51.005+0000 [id=51] INFO hudson.util.Retrier#sta
```

Access Jenkins

URL:

<http://192.168.0.150:8080>

```
root@nagios:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP>
    link/loopback 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP>
    link/ether 08:00:27:4c:00:00
    altname enx0800274ce86600
    inet 192.168.0.150/22 brd 192.168.0.255 scope global dynamic nopromisc
        valid_lft forever preferred_lft forever
```

The screenshot shows a web browser window with the address bar displaying 'Not secure 192.168.0.150:8080'. The page title is 'Getting Started' and the main heading is 'Create First Admin User'. The form contains four input fields: 'Username', 'Password', 'Confirm password', and 'Full name'. At the bottom of the form, there is a 'Jenkins 2.440.3' label, a 'Skip and continue as admin' link, and a 'Save and Continue' button.

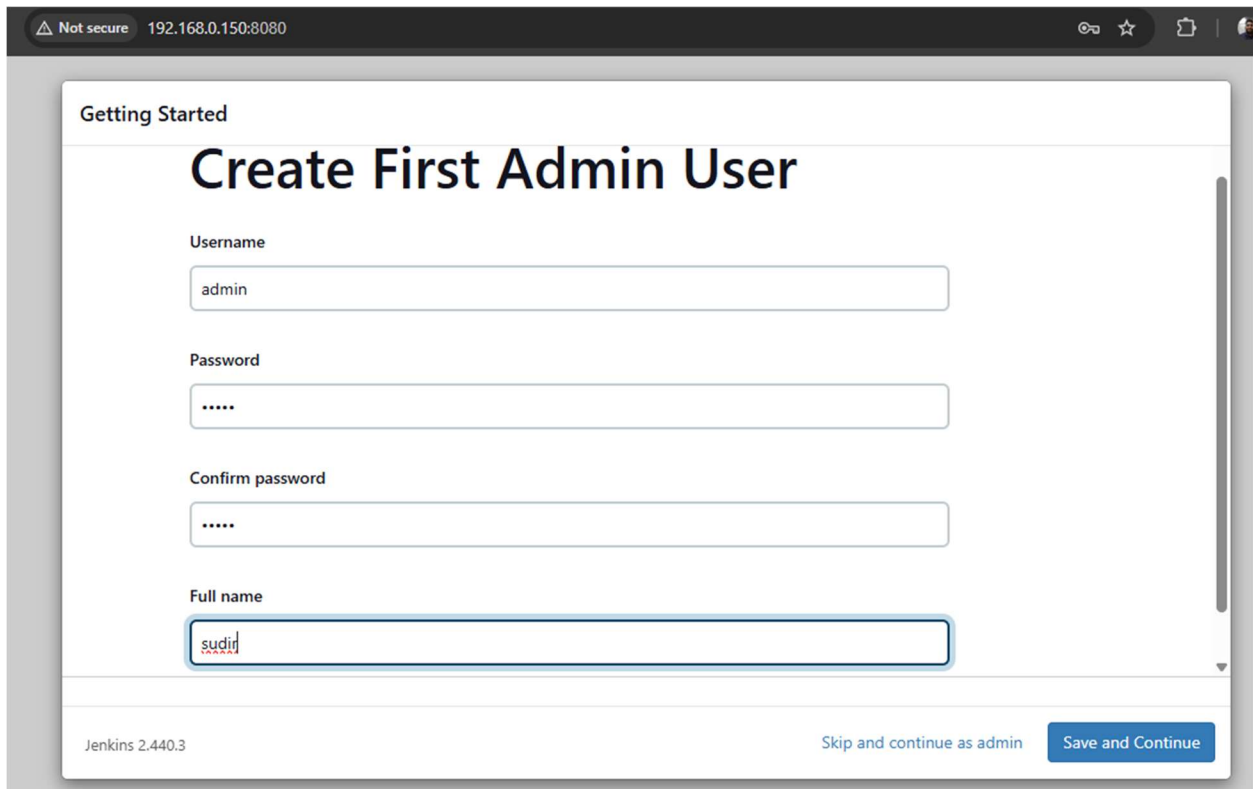
Unlock Password:

`sudo cat /var/lib/jenkins/secrets/initialAdminPassword`

```
root@nagios:~# cat /var/lib/jenkins/secrets/initialAdminPassword
91377cd31ced1a1c
```

Jenkins Dashboard

After installing plugins and creating admin user.



The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is 192.168.0.150:8080. The page title is 'Getting Started'. The main heading is 'Create First Admin User'. There are four input fields: 'Username' with the value 'admin', 'Password' with masked characters '.....', 'Confirm password' with masked characters '.....', and 'Full name' with the value 'sudir'. At the bottom left, it says 'Jenkins 2.440.3'. At the bottom right, there are two buttons: 'Skip and continue as admin' and 'Save and Continue'.

Getting Started

Create First Admin User

Username

Password

Confirm password

Full name

Jenkins 2.440.3

Skip and continue as admin

Save and Continue

Job Creation

Created a freestyle job.

Build Script:

```
echo "Sudir jenkins task 1"
```

```
date
```

Dashboard > jenkins task >

Status

</> Changes

Workspace

Build Now

Configure

Delete Project

Rename

jenkins task

task 1

Permalinks

- Last build (#1), 41 sec ago
- Last stable build (#1), 41 sec ago
- Last successful build (#1), 41 sec ago
- Last completed build (#1), 41 sec ago

Build History

trend

Filter...

Build Execution

Job executed successfully.

#1 (Apr 8, 2026, 11:49:50 AM)

Keep this build forever

Add description

Started 1 min 46 sec ago

Took 0.29 sec

</>

No changes.

Started by user sudir

Console Output

Dashboard > jenkins task > #1 > Console Output

Status

</> Changes

Console Output

View as plain text

Edit Build Information

Delete build '#1'

Console Output

Started by user sudir

Running as SYSTEM

Building in workspace /var/lib/jenkins/workspace/jenkins task

[jenkins task] \$ /bin/sh -xe /tmp/jenkins4570918279904738250.sh

+ echo Sudir jenkins task 1

Sudir jenkins task 1

+ date

Wed Apr 8 11:49:51 AM UTC 2026

Finished: SUCCESS

User Management

Created users via Jenkins UI.

Proof: